Certified Coverage of Exponential Search Spaces

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Idea. Consider a search problem whose naive state space has size 2^n. Some states may admit a compact structural representation; others may be proposed for elimination by a learned model and certified exactly; still others may be represented by exact summaries of paired groups. Only the residual must be searched explicitly. The central quantity is the *growth of what remains uncovered*, together with the cost of obtaining coverage.

A coverage model

Let $\mathcal{S}_n$ be a nominal search space with

$$|\mathcal{S}_n| = 2^n.$$

We use three mechanisms to cover disjoint parts of this space.

First, a structural procedure identifies a set $K_n \subseteq \mathcal{S}_n$ of size

$$k(n) = |K_n|$$

that admits a compact representation. In the simplest logarithmic model, the $k(n)$ explicit states are replaced by a description of cost

$$C_{\text{comp}}(n) = \log_2 k(n).$$

This is deliberately an abstract assumption: it describes the cost once such a representation is available, not a universal property of arbitrary subsets.

Second, a learned classifier proposes regions of the remaining space for removal. Let $\delta(n)$ denote the total number of states covered by these proposals. The classifier is not trusted to delete states directly. Instead, an exact verifier certifies proposed regions. If $a(n) \in [0, 1]$ is the fraction of proposed mass that is successfully certified, then the safely eliminated mass is

$$\delta_c(n) = a(n)\,\delta(n).$$

Prediction error therefore affects efficiency, not correctness: uncertified proposals simply remain in the residual problem.

Third, pair groups of states and replace each pair by an exact summary for the search question. Let $p(n)$ count the additional states covered by such summaries, excluding states already counted in $k(n)$ or $a(n)\delta(n)$. Let $Q_{\text{pair}}(n)$ include the cost of finding pairs, constructing and certifying their summaries, and using the resulting representation.

The explicit residual is

$$r(n) = 2^n - k(n) - a(n)\delta(n) - p(n). \tag{1}$$

Let $P(n)$ collect the cost of discovering the structural representation, running the learned proposal mechanism, certifying its proposed regions, and managing the resulting representation. Under the simple assumption that explicit residual search costs $r(n)$ units, the total modeled cost is

$$T(n) = P(n) + Q_{\text{pair}}(n) + r(n) + \log_2 k(n). \tag{2}$$

Thus

$$\begin{aligned} T(n) = P(n) + Q_{\text{pair}}(n) + 2^n - k(n) \\ - a(n)\delta(n) - p(n) + \log_2 k(n). \end{aligned} \tag{3}$$

The separation is important. Large covered regions help only if discovering, representing, and certifying them is cheap.

The certified coverage theorem

Define the total *certified coverage*

$$C(n) = k(n) + a(n)\delta(n) + p(n),$$

so that $r(n) = 2^n - C(n)$.

Theorem 1 (Certified coverage criterion). Assume $P(n) + Q_{\text{pair}}(n) = \text{poly}(n)$, $1 \le k(n) \le 2^n$, and the residual is searched in time proportional to $r(n)$. Then

$$T(n) = \text{poly}(n) \quad \Longleftrightarrow \quad 2^n - C(n) = \text{poly}(n).$$

Equivalently, polynomial modeled runtime occurs exactly when

$$C(n) = 2^n - \text{poly}(n).$$

Proof. From (2),

$$T(n) = P(n) + Q_{\text{pair}}(n) + \big(2^n - C(n)\big) + \log_2 k(n).$$

Because $k(n) \le 2^n$,

$$0 \le \log_2 k(n) \le n.$$

The overhead and logarithmic terms are polynomially bounded. Hence $T(n)$ is polynomial exactly when the residual $2^n - C(n)$ is polynomially bounded. □

The theorem gives a sharp asymptotic warning: eliminating a very large *percentage* of an exponential space need not change its asymptotic order.

Corollary 1 (Fixed uncovered fraction). If there exists a constant $\varepsilon > 0$ such that, for infinitely many n,

$$C(n) \le (1 - \varepsilon)2^n,$$

then

$$r(n) \ge \varepsilon 2^n$$

and therefore, along that infinite subsequence, the modeled runtime satisfies

$$T(n) = \Omega(2^n)$$

Thus 90%, 99%, or even 99.999% coverage is asymptotically insufficient if the uncovered fraction stays bounded away from zero. Polynomial behavior requires

$$\frac{C(n)}{2^n} = 1 - \frac{\mathrm{poly}(n)}{2^n},$$

so the uncovered fraction must vanish exponentially fast, up to polynomial factors.

Differential form

The discrete complexity statement has a useful continuous interpolation. Treat n as a continuous scale parameter and differentiate (1):

$$\frac{dr}{dn} = (\ln 2)2^n - k'(n) - a(n)\delta'(n) - a'(n)\delta(n) - p'(n). \tag{4}$$

The terms are raw state creation, structural coverage, new certified proposals, improving certification rate, and paired coverage, respectively.

Differentiating the full cost gives

$$\begin{aligned}\frac{dT}{dn} = (\ln 2)2^n - \left(1 - \frac{1}{k\ln 2}\right) k' - a\delta' - a'\delta - p' \\ + P'(n) + Q'_{\mathrm{pair}}(n).\end{aligned} \tag{5}$$

For large k, the $1/(k\ln 2)$ correction is negligible, giving the approximate balance law

$$\begin{aligned}T'(n) \approx (\ln 2)2^n - k' - a\delta' - a'\delta - p' \\ + P'(n) + Q'_{\mathrm{pair}}(n).\end{aligned} \tag{6}$$

Equivalently, since $r(n) = 2^n - C(n)$,

$$r'(n) = (\ln 2)2^n - C'(n).$$

Integrating from a reference scale n_0,

$$r(n) = r(n_0) + \int_{n_0}^{n} \left[(\ln 2)2^s - C'(s)\right] ds. \tag{7}$$

This form makes the central picture especially transparent: polynomial behavior requires the cumulative gap between exponential state creation and certified coverage growth to remain only polynomial in n.

Why certification matters

A naive learned-pruning model might discard $\delta(n)$ states with error rate $e(n)$. If every erroneous deletion has to be recovered to preserve exactness, an error residue of order $e(n)\delta(n)$ appears. When $\delta(n) = \Theta(2^n)$, any fixed $e > 0$ leaves an exponential residue. To keep that residue polynomial would require

$$e(n)\,2^n = \mathrm{poly}(n), \qquad \text{hence} \qquad e(n) \le \frac{\mathrm{poly}(n)}{2^n}.$$

Thus ordinary constant-error prediction is not enough for exact bulk deletion of an exponential region.

Certification avoids that failure mode. The learned model becomes a *proposal mechanism*: it suggests large regions that may be irrelevant, while an exact verifier alone has authority to delete them. A poor classifier can waste computation, but it cannot make the final algorithm incorrect. In this formulation, learning changes δ and the acceptance function a; correctness is carried by verification.

The cost of pairing

For disjoint groups B_0, B_1, a summary s must support an exact merge rule

$$s(B_0 \cup B_1) = \mathrm{Merge}\big(s(B_0), s(B_1)\big),$$

and the final summary must answer the original search question exactly. A complete explicit pairing tree over 2^n states has $2^n - 1$ merges, even though only one group remains. Thus pairing helps asymptotically only when a compact rule or reuse of summaries keeps $Q_{\mathrm{pair}}(n)$ polynomial. Counting covered states without this cost would hide the original exponential work.

Scope and the remaining difficulty

The coverage criterion is a theorem about the model; it does not assert that every concrete search problem admits such coverage. The hard question is whether, for a given problem, one can simultaneously achieve

$$P(n) + Q_{\mathrm{pair}}(n) = \mathrm{poly}(n) \qquad \text{and} \qquad 2^n - C(n) = \mathrm{poly}(n).$$

Cardinality alone cannot answer this: exponentially many states can sometimes be described by a short rule. The substantive lower- or upper-bound problem is therefore about the *cost of discovering and certifying structure*, not merely the number of states represented by that structure.

General representation cost

The logarithmic term is the motivating special case, not an essential part of the coverage theorem. If a compressed region of size $k(n)$ has representation cost $\rho(n, k)$, write

$$T(n) = P(n) + Q_{\mathrm{pair}}(n) + r(n) + \rho(n, k(n)).$$

Whenever $P(n) + Q_{\mathrm{pair}}(n)$ and $\rho(n, k(n))$ are polynomially bounded, the same proof gives

$$T(n) = \mathrm{poly}(n) \quad \Longleftrightarrow \quad r(n) = 2^n - C(n) = \mathrm{poly}(n).$$

Thus the essential object is the uncovered residual; $\log_2 k$ is simply the strongest compression regime considered here. This also separates two questions that should not be conflated: how many states a representation covers, and how costly that representation is to discover and manipulate.

Summary. Exponential search becomes a competition between state-space growth and certified coverage. Structural compression and exact pairing represent states collectively; learning proposes further regions; certification makes eliminations exact; and the residual determines explicit work. The governing condition is simple:

$$\text{polynomial overhead} + \text{polynomial residual} \implies \text{polynomial modeled runtime}.$$

The differential form records the same statement dynamically: coverage must track exponential state creation closely enough that their accumulated gap is only polynomial.